Energy Economics 40 (2013) 207–221

Contents lists available at ScienceDirect

Energy Economics

journal homepage: www.elsevier.com/locate/eneco

Forecasting carbon futures volatility using GARCH models with energy volatilities

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ARTICLE INFO

Article history:
Received 16 April 2012
Received in revised form 10 June 2013
Accepted 28 June 2013
Available online 10 July 2013

JEL classification:
C22
G15
G17
Q47
Q54

Keywords:
Carbon futures
GARCH
Implied volatility
Forecasting
Energy market

ABSTRACT

This article examines the volatility forecasting abilities of three approaches: GARCH-type model that uses carbon futures prices, an implied volatility from carbon options prices, and the *k*-nearest neighbor model. Based on the results, we document that GARCH-type models perform better than an implied volatility and the k-nearest neighbor model. This result suggests that carbon options have little information about carbon futures due to their low trading volume. We also investigate whether the volatilities of energy markets, i.e., Brent oil, coal, natural gas, and electricity, forecast following day's carbon futures volatility. According to the results, we suggest that Brent oil, coal, and electricity may be used to forecast the volatility of carbon futures.

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1. Introduction

GARCH-type models are widely used to estimate volatility of financial asset returns and show good performance. However, according to the survey results of Poon and Granger (2003), GARCH-type models are not necessarily the best forecasting model, as they give satisfactory results for in-sample forecasts, but not for out-of-sample forecast estimation. Generally, there are two alternatives to GARCH-type models: historical volatility and implied volatility (IV). The historical volatility approach predicts volatility using the past volatility of a sample. The random walk, historical average, simple moving average, exponential smoothing, exponentially weighted moving average, and *k*-nearest neighbor (k-NN) methods all belong to this approach. Apart from random walk and historical average, successful applications of the historical volatility approach normally involve searching for the optimal lag length or weighting scheme in an estimation period for out-of-sample forecasting. The second alternative to GARCH-type models, IV, uses the Black–Scholes option pricing formula. In this formula, all parameters other than volatility are observed in financial markets. After inserting these parameters into the Black–Scholes formula, volatility is the only unknown parameter. The formula is then inverted to derive an estimate of volatility implied by the observed option price. Eventually, an estimate of volatility becomes IV.

The choice of method to forecast returns volatility depends on comparing the performance of several alternatives. Poon and Granger (2003) point out that, in a forecasting exercise, comparing the forecasting performance of the different methods is most important. There are several approaches to assess the performance of forecasts. These approaches comprise computing the difference between the forecasts and the observed value, running regression to investigate the information content of forecasts, and measuring the relative usefulness of forecasts on the basis of a utility function or in relation to common uses of volatility forecasts in financial markets (Agnolucci, 2009). This article focuses on the first two of these approaches. Agnolucci (2009) indicates that these two approaches are traditional in financial econometric literature.

Although there are many studies on most financial assets, much remains to be done in the carbon market. The carbon market has been active under the EU Emission Trading Scheme (ETS) since 2005, after the Kyoto Protocol came into force. Under the EU ETS, spot, futures, and options of the EU allowance (EUA; the legal aspect) and Certified

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0140-9883/$ – see front matter © 2013 Elsevier B.V. All rights reserved.
http://dx.doi.org/10.1016/j.eneco.2013.06.017

Emission Reduction (CER; the voluntary aspect) are traded, with the trading volume of the EUA being larger than that of CER.[1] Following the Kyoto Protocol, phase I was from 2005 to 2007, phase II from 2008 to 2012, and phase III is from 2013 to 2020. Following Daskalakis et al. (2009), there was a market correction which resulted from the inter-phase banking restriction and the over-allocation problem. However, since the advent of phase II, this restriction has moderated and the amount of the allocation has been adjusted (Chevallier, 2011b). Moreover, the trading volume and market evolution have grown since 2005.[2] Therefore, the carbon market is moving to mature state, and the econometric application to the carbon market is a valuable exercise. Here, we focus only on EUA futures in phase II because of its high trading volume.

The purpose of this article is to examine the predictive power of GARCH-type, IV, and k-NN models of EUA futures returns in phase II. The predictive power of each is assessed using various loss functions, such as the mean square error (MSE), MSE-LOG, mean absolute error (MAE), MAE-LOG, and QLIKE, as well as the regression-based approach, the Diebold–Mariano test, the superior predictive ability test, and the model confidence set. In GARCH-type models, we use various types of GARCH models and error distributions. Another contribution of this article is to investigate whether EUA futures volatility is related to energy market volatility. In contrast to previous literature, for example Chevallier (2009, 2010), and Chevallier and Sévi (2011), we first assess the relationship between EUA futures volatility and energy market volatility using the linear regression approach.

This article is organized as follows. Section 2 details the methodology used in this article and discusses previous literature. Section 3 describes the data used. Section 4 presents the results from the GARCH-type models estimation. Section 5 contains the results from the model comparison. Section 6 describes the results using GARCH-type model with energy volatilities. Section 7 concludes this article.

2. Methodology and literature review

To forecast the volatility of financial time series, various GARCH-type models have been used. The traditional GARCH(p,q) model, where p is the order of the moving average ARCH term and q is the order of the autoregressive GARCH term, can be written as (Bollerslev, 1986):

$$A(L)y_t = X_t'\theta + B(L)\varepsilon_t, \tag{1}$$

$$\varepsilon_t = h_t e_t, \tag{2}$$

$$h_t^2 = \omega + \sum_{i=1}^{p}\alpha_i\varepsilon_{t-i}^2 + \sum_{j=0}^{q}\beta_j h_{t-j}^2, \tag{3}$$

where the two terms $A(L)$ and $B(L)$ in Eq. (1) are two polynomials of lags, and X_t is a vector of exogenous variables. Using Eq. (3) in this simplest model, the current conditional variance is forecasted as the function of a constant term, ω, the ARCH term, ε_{t-i}^2, and the GARCH term, h_{t-j}^2. More complicated GARCH-type models improve Eq. (3) to explain the asymmetric effect. Following Chevallier (2009), GARCH-type models with the asymmetric effect provide the best fit to the carbon futures prices among other GARCH-type specifications. Therefore, we use GARCH-type models with the asymmetric effect to estimate the parameters of the volatility equation. Nelson (1991) proposes the Exponential GARCH (EGARCH) model as:

$$\log h_t^2 = \omega + \sum_{i=1}^{p}\alpha_i\left(\frac{\varepsilon_{t-i}}{h_{t-i}}\right) + \sum_{j=0}^{q}\beta_j \log h_{t-j}^2 + \sum_{i=1}^{p}\gamma_i\left[\left|\frac{\varepsilon_{t-i}}{h_{t-i}}\right| - E\left(\left|\frac{\varepsilon_{t-i}}{h_{t-i}}\right|\right)\right], \tag{4}$$

where γ_i shows the presence of the asymmetric effect. The Threshold GARCH (TGARCH) model by Zakoian (1994) can be written as:

$$h_t = \omega + \sum_{i=1}^{p}\left(\alpha_i^+\varepsilon_{t-i}^+ - \alpha_i^-\varepsilon_{t-i}^-\right) + \sum_{j=0}^{q}\beta_j h_{t-j}, \tag{5}$$

where $\varepsilon_{t-i}^+ = \max(\varepsilon_{t-i},0)$ and $\varepsilon_{t-i}^- = \min(\varepsilon_{t-i},0)$, i.e., the positive and negative parts of ε_{t-i}, respectively. Glosten et al. (1993) introduce the GJR-GARCH model as follows:

$$h_t^2 = \omega + \sum_{i=1}^{p}\alpha_i\varepsilon_{t-i}^2 + \sum_{j=0}^{q}\beta_j h_{t-j}^2 + \sum_{i=1}^{p}\gamma_i\varepsilon_{t-i}^2\Gamma_{t-i}, \tag{6}$$

where $\Gamma_{t-i} = 1$ if $\varepsilon_{t-i} < 0$, and 0 otherwise. In the TGARCH and GJR-GARCH models, negative or positive values of ε_{t-i} indicate bad or good news, respectively. Although Chevallier (2011d), Feng et al. (2011), and Chevallier and Sévi (2011) do not use GARCH models, instead using non-parametric modeling, nonlinear dynamics, and the HAR-RV model, respectively, several studies have employed GARCH-type models to estimate carbon volatility. Paolella and Taschini (2008) and Benz and Trück (2009) use an AR(1)-GARCH(1,1) model to fit EUA spot return dynamics. Chevallier (2010) also uses an AR(1)-GARCH(1,1) model to estimate the volatility of EUA spot and futures. In Chevallier (2009), various types of GARCH models, including GARCH, EGARCH, and TGARCH, are employed to estimate EUA futures volatility. Mansanet-Bataller et al. (2011) use TGARCH(1,1) to fit the VAR(4) residuals of EUA futures, and Chevallier (2011b) detects volatility instability using an EGARCH(1,1) model.

Implied volatility (IV) measures the expected volatility of an underlying asset of options in the future. The simplest method to compute the IV of options on futures comes from Black pricing formula (Black, 1976). According to this formula, IV can be obtained by inverting $C = [FN(d_1) - KN(d_2)]e^{-rT}$ where C is the call price, F is the current futures price, T is the time to expiration of the call option, K is the strike price of the call option, r is the risk-free interest rate, and N refers to the cumulative normal distribution values at $d_1 = [\ln(F/K) + (\sigma^2/2)T]/\sigma\sqrt{T}$ and $d_2 = d_1 - \sigma\sqrt{T}$. The value of σ indicates the implied volatility of the underlying futures. We use this approach following Agnolucci (2009) and Chevallier (2011b). Chevallier (2011b) calculates the IV of carbon futures by minimizing the difference between real call option prices and Black call option prices.

In econometrics, the k-nearest neighbor method is used to estimate the density function of observed time series data (Silverman, 1986), or predict future value of given time series data. According to Clements and Hendry (2011), Farmer and Sidorowich (1987) introduce this method to the field of forecasting. When the k-nearest neighbor method predicts the volatility, this method uses the block values of previous volatility data. We consider a block of length m from the observed time series y_t, $t = 1,\cdots,T$, and denote it by $\mathbf{y}_t^m = (y_t, y_{t-1}, \cdots, y_{t-(m-1)})$. Therefore, there are $T - m + 1$ blocks. Then, according to the distance measure $d(\mathbf{y}_T^m, \mathbf{y}_t^m)$, we carry out a search to find the k earlier blocks that are closest to $\mathbf{y}_T^m$. The simplest distance measure is defined as $d(\mathbf{y}_T^m, \mathbf{y}_t^m) = \sum_{i=0}^{m-1}|y_{T-i} - y_{t-i}|$, and the k earlier blocks are the k nearest

[1] We can find this fact in several carbon trading exchanges, e.g., European Climate Exchange (ECX), Nordpool, European Energy Exchange (EEX), Powernext, and so on.

[2] ECX reports historical monthly trading volumes on their official website. According to ECX, the yearly trading volume in 2011 was approximately 71 times more than the trading volume in 2005. The World Bank reports carbon market evolution every year, and shows that the market evolution of 119.8 billion dollars in 2010 is about 15 times the 7.9 billion dollars in 2005.

neighbors $\mathbf{y}_{t_1}^m, \cdots \mathbf{y}_{t_k}^m$ of $\mathbf{y}_T^m$. With these nearest neighbors, we run the regressions as follows:

$$y_{t_i+1} = \alpha + \sum_{j=0}^{m-1} \beta_j y_{t_i-j} + \varepsilon_{t_i+1}, \quad i = 1, \cdots, k, \tag{7}$$

$$\widehat{y}_{Forecast,T} = \alpha' + \sum_{j=1}^{m} \beta_j{}' y_{T+1-j}, \tag{8}$$

where $\hat{y}_{Forecast,t}$ is the one-day-ahead forecast value at T, and α' and β' are the estimated value in Eq. (7).

To assess the predictive power of the volatility forecasts from GARCH-type, IV, and k-NN models, we compute the value of various loss functions for each model, i.e., the mean square error (MSE), MSE-LOG, mean absolute error (MAE), MAE-LOG, and QLIKE. These statistics are used to compare the predictability of various volatility forecast models (Hansen and Lunde (2006), Patton and Sheppard (2009), and Patton (2011)). We use the definition of the QLIKE following Patton (2011). When we define y_{t+1} and $\hat{y}_{t,1}$ as the volatility observed between t and $t+1$ and one period volatility forecast at time t, respectively, MSE, MSE-LOG, MAE, MAE-LOG, and QLIKE can be obtained as follows:

$$MSE = \sum_{t=1}^{N} \left(\hat{y}_{t,1} - y_{t+1}\right)^2 / N, \tag{9}$$

$$MSE-LOG = \sum_{t=1}^{N} \left(\log \widehat{y}_{t,1} - \log y_{t+1}\right)^2 / N, \tag{10}$$

$$MAE = \sum_{t=1}^{N} \left|\hat{y}_{t,1} - y_{t+1}\right| / N, \tag{11}$$

$$MAE-LOG = \sum_{t=1}^{N} \left|\log \hat{y}_{t,1} - \log y_{t+1}\right| / N, \tag{12}$$

$$QLIKE = \sum_{t=1}^{N} \left(\log y_{t+1} + \frac{\hat{y}_{t,1}}{y_{t+1}}\right) / N. \tag{13}$$

Furthermore, following Agnolucci (2009), we assess the predictive power of the volatility forecasts by using regression-based approaches. The information content of the volatility forecasts can be evaluated on the basis of the parameter β in

$$y_{t+1} = \alpha + \beta \hat{y}_{t,1} + \nu_{t+1}, \tag{14}$$

which should be statistically significant and greater than zero. The values of R^2 and the information criteria, i.e., the Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC) of Eq. (14), can be used as the basis for evaluating information content. The unbiasedness of the model is examined by testing the hypothesis that $\alpha = 0$ and $\beta = 1$. In fact, Eq. (14) comes from Mincer and Zarnowitz (1969), and is called the Mincer–Zarnowitz regression. To evaluate relative information content, we use combined regression, as follows:

$$y_{t+1} = \alpha + \beta_1 \hat{y}_{1t,1} + \beta_2 \hat{y}_{2t,1} + \nu_{t+1}, \tag{15}$$

where $\hat{y}_{1t,1}$ and $\hat{y}_{2t,1}$ are the forecasts from two alternative models. If $\widehat{y}_{1t,1}$ is an information efficient forecast, β_2 should be statistically insignificant and the values of R^2 and the information criteria should not be remarkably different from those of Eq. (14) with $\hat{y}_{1t,1}$ only. Otherwise, the combined model of Eq. (15) may be used to forecast the carbon futures volatility.

To evaluate relative predicting accuracy of volatility forecasts, we also use the Diebold–Mariano test statistic of Diebold and Mariano (1995), the superior predictive ability test, and the model confidence set. The Diebold–Mariano test is based on the loss function differential between model i and j, which is defined as $d_{i,j,t} = L\left(y_{t+1}, \hat{y}_{t,1}^i\right) - L\left(y_{t+1}, \hat{y}_{t,1}^j\right)$, where $L(\ )$ means a loss function. The null hypothesis of equal predictive accuracy is then $H_0 : E[d_{i,j,t}] = 0$, and the Diebold–Mariano test statistic takes the form of a t-statistic, i.e.,

$$DM_T = \frac{\overline{d}_{i,j}}{\left(asy.var\left(\overline{d}_{i,j}\right)\right)^{1/2}} \overset{a}{\sim} N(0,1), \tag{16}$$

where $\overline{d}_{i,j} = N^{-1} \sum_{t=1}^{N} d_{i,j,t}$ and $asy.var\left(\overline{d}_{i,j}\right)$ is the asymptotic variance of $\overline{d}_{i,j}$. Therefore, by comparing the Diebold–Mariano test statistic and the critical value, we reject or accept the null hypothesis. While the Diebold–Mariano test is the approach to pairwise comparison of two forecast models, the superior predictive ability test and the model confidence set are the approach to multiple comparisons of alternative forecasts to the benchmark forecast model. Following Hansen (2005), the null hypothesis of the superior predictive ability test is $H_0 : E[\mathbf{d}_t] \le \mathbf{0}$ where $\mathbf{d}_t = (d_{1,t}, \cdots, d_{m,t})$, $d_{j,t} = L\left(y_{t+1}, \hat{y}_{t,1}^0\right) - L\left(y_{t+1}, \hat{y}_{t,1}^j\right)$, and $\hat{y}_{t,1}^0$ is the benchmark forecast model. The superior predictive ability test statistic takes the form of

$$SPA_T = \max\left[\max_k \frac{\overline{d}_k}{\left(asy.var\left(\overline{d}_k\right)\right)^{1/2}}, 0\right], \tag{17}$$

and the null distribution of the SPA_T is based on $\sqrt{N}\overline{\mathbf{d}} \to^d N\left(\hat{\boldsymbol{\mu}}^c, \hat{\Omega}\right)$, where $\hat{\boldsymbol{\mu}}^c$ is a consistent estimator of $\boldsymbol{\mu} = E[\mathbf{d}_t]$, and $\hat{\Omega}$ is a consistent estimator of the covariance matrix of $\overline{\mathbf{d}}$. Then, using a bootstrap implementation, we seek the distribution of the test statistic under the null hypothesis and the critical value, and reject or accept the null hypothesis. According to Hansen et al. (2011), the model confidence set test involves a sequence of tests for equal predictive ability. Given a set M^0, which contains m forecast models, the null hypothesis is that all models in M^0 have equal predictive accuracy, and is described as $H_0 : E[d_{i,j,t}] = 0\ \forall i, j \in M^0$. If the null hypothesis is rejected at a given confidence level α, then the worst performing model in M^0 is eliminated. After that, the equal predictive accuracy test is repeated until the null hypothesis is accepted. When the null hypothesis is accepted, the remainder composes $1 - \alpha$ confidence set, $\hat{M}_{1-\alpha}^*$. Hansen et al. (2011) provide the two statistics and a detailed description of the bootstrap implementation. The two statistics, used in the model confidence set test, are expressed as follows:

$$MCS_T_{\max,M} = \max_{i \in M} t_{i,\cdot}, \tag{18}$$

$$MCS_T_{R,M} = \max_{i,j \in M} \left|t_{i,j}\right|, \tag{19}$$

where $t_{i,\cdot} = \frac{\overline{d}_{i,\cdot}}{\left(asy.var\left(\overline{d}_{i,\cdot}\right)\right)^{1/2}}$, $t_{i,j} = \frac{\overline{d}_{i,j}}{\left(asy.var\left(\overline{d}_{i,j}\right)\right)^{1/2}}$, and $\overline{d}_{i,\cdot} = m^{-1} \sum_{j \in M} \overline{d}_{i,j}$ for $i, j \in M$.

Financial markets are usually correlated with each other. To examine whether one market is correlated with other markets, the vector error correction model (VECM) with a cointegration test, the Markov-switching vector autoregression (VAR) model, and linear regression analysis are employed. In the case of the carbon market, several studies report that the carbon market is involved in energy markets or the macro economy.

With respect to the VECM with cointegration test, Fezzi and Bunn (2009) show that electricity and natural gas markets in the UK interact with the EUA spot market. According to them, the natural gas price influences the EUA spot price, and natural gas and EUA spot prices affect the electricity price. In particular, a 1% increase in the natural gas (carbon

spot) price is associated with an electricity price rise of 0.66% (0.32%). Bredin and Muckley (2011) point out that the EUA futures price is correlated with the Brent oil price, energy spreads, and the equity index. From their result that significant cointegration factors in Phase II are different from those in Phase I, they suggest that a new pricing regime of EUA futures in Phase II arises. Chevallier (2011c) examines the relationship between EUA futures prices and industrial production, i.e., the macro economy, using two-regime threshold VECM. According to Chevallier (2011c), EUA futures prices are affected by the lagged one period industrial production index.

Chevallier (2011a, 2011c) use the Markov-switching VAR model. Chevallier (2011a) uses the model to examine whether macroeconomic activity, i.e., the EU industrial production index and energy prices, has an effect on the price of EUA futures. The results show that the EU industrial production index has a positive (negative) impact on the carbon market during a period of economic expansion (recession). Moreover, Chevallier (2011a) suggests that the carbon market is related to Brent oil and natural gas, but not coal. According to Chevallier (2011c), the EU industrial production index affects EUA futures prices with a lag, and this phenomenon seems to result from the institutional constraints of the carbon market.

Finally, several studies use the linear regression analysis approach to investigate the relation between the carbon market and other markets. Mansanet-Bataller et al. (2007) use this approach to show that the EUA futures index return is related to Brent oil and natural gas returns. Alberola et al. (2008), Daskalakis and Markellos (2009), Oberndorfer (2009), Hintermann (2010), and Mansanet-Bataller et al. (2011) also use linear regression. Alberola et al. (2008) show that the EUA price is related to the prices of Brent oil, natural gas, coal, electricity, and energy spreads. Daskalakis and Markellos (2009) find a positive relationship between the EUA spot return and electricity risk premia through linear regression analysis. Oberndorfer (2009) shows that EUA spot price changes are positively correlated with EU electricity firms' stock returns. According to Oberndorfer (2009), the volatility of EUA spot prices seems to have no significant relationship with electricity firms' stock returns. Hintermann (2010) shows that, from April 2006 to June 2007, natural gas and coal prices had an impact on the EUA spot price. Mansanet-Bataller et al. (2011) suggest that the EUA futures price is affected by Brent oil, coal, and natural gas.

Previous literature focuses on the relationship between carbon price and other markets. However, according to Guo and Kliesen (2005), it is possible to investigate whether the volatility of one market is affected by the other market's volatility using the linear regression approach. This approach can easily examine the direct relationship between the volatilities of several markets. Therefore, this approach is the most powerful and simplest method to investigate the volatility relationship across various markets. In this article, we use this linear regression analysis to examine whether carbon futures volatility is affected by energy market volatilities, i.e., the volatilities of Brent oil, coal, natural gas, and electricity. To make the forecast containing energy market volatilities, we use the following linear regressions:

$$y_{t+1} = \alpha + \beta_{Forecast}\hat{y}_{Forecast,t} + \nu_{t+1}, \tag{20}$$

$$\hat{y}_{Forecast,t} = \alpha' + \beta_1'\hat{y}_{GARCH,t} + \beta_2' y_{Brent,t} + \beta_3' y_{gas,t} + \beta_4' y_{coal,t} + \beta_5' y_{electricity,t}, \tag{21}$$

$$y_{in,t+1} = \alpha' + \beta_1'\hat{y}_{GARCH,t} + \beta_2' y_{Brent,t} + \beta_3' y_{gas,t} + \beta_4' y_{coal,t} + \beta_5' y_{electricity,t} + \nu_{in,t+1}, \tag{22}$$

where $\hat{y}_{GARCH,t}$ is the best one period forecast among GARCH-type models, IV, and k-NN models at time t, and $y_{Brent,t}$, $y_{gas,t}$, $y_{coal,t}$, and $y_{electricity,t}$ are the volatilities of Brent oil, natural gas, coal, and electricity observed between time $t-1$ and t, respectively. In Eq. (21), α', β_1', β_2', β_3', β_4', and β_5' are estimated in Eq. (22), which indicates in-sample period estimation.

3. Data

We use daily and intraday data for the European Climate Exchange (ECX)[3] carbon futures contracts of maturity from December 2008 to December 2011 to estimate GARCH models. The data set runs from January 03, 2008 to August 15, 2011[4], i.e., a time period that covers approximately three and a half years from the start of Phase II. The sample size totals to 922 observations based on the daily data. Daily returns are computed by taking the natural logarithm of the ratio of two successive days' closing prices, i.e., $r_t = \ln(p_t/p_{t-1})$. The trading unit of carbon futures is €/tCO_2e.[5]

Parameters for the GARCH-type models used in the forecast are estimated on a 452-day rolling sample, i.e., using a window that shifts forward as new information arrives. When estimating a GARCH model using a 452-day rolling sample from January 03, 2008 to October 08, 2009, we can obtain a one-day-ahead forecast for October 09, 2009. By shifting the estimation window by one day and re-estimating the model over the new sample, we can obtain a forecast for October 10, 2009. This estimation process is repeated until we obtain a forecast for August 15, 2011.

The IV data of ECX carbon futures options used in this study are obtained from the ECX quoted data. ECX computes the value of IV using Black formula with a 6-month Euribor swap rate as the risk-free interest rate. Following Agnolucci (2009), we use the values of IV for the nearest-to-the-money options of the nearby series and then compute the arithmetic average of both call and put options, respectively. The nearest-to-the-money options[6] indicate the options with strike prices above and below the underlying price. The nearby series is defined as the series with the shortest time to expiration but with maturity of at least eight calendar days. The quoted IV is based on the remaining year to the maturity. Therefore, we divide the values of IV by the square root of 252 (the average number of business days in one year), and then divide again by the number of days to expiration.

For the k-NN method, we assume that the squared daily log returns at t are the variance values at time t, following Agnolucci (2009). In other words, we use the absolute values of daily log returns of carbon futures as the volatility time series. The block size, m, and the number of nearest neighbor, k, are defined as 6 and 9. The criterion of specifying these numbers is shown in Bajo-Rubio et al. (2001).

As volatility is a latent variable, a proxy for volatility has to be computed to be compared to the forecasts. We use the realized volatility from the intraday data of ECX carbon futures contracts as a volatility proxy, i.e., y_{t+1} in Eqs. (14), (15), and (20). When we divide one trading day into $1/m$ m-minute periods and denote an m-minute log return at t as $r_{t,m} = \ln(p_t/p_{t-m})$, the realized volatility within one trading day is computed as follows:

$$RV = \sqrt{\sum_{j=1}^{1/m} r_{t+jm,m}^2}. \tag{23}$$

We use the intraday data of ECX carbon futures contracts from October 09, 2009 to August 15, 2011. The daily transaction record begins at 07:00 (GMT), and ends at 17:00 (GMT). Fig. 1 shows the volatility signature plot for the period of out-of-sample forecast. Following

[3] Under EU ETS, ECX has the largest trading volume of carbon derivatives among carbon trading markets.

[4] The expiration is switched to the second-month contract on the last Thursday of the nearby month contract.

[5] 1tCO_2e indicates one ton of CO_2 equivalent, and is a basic unit for carbon emission allowance.

[6] Agnolucci (2009) documents the advantages of using the nearest-to-the-money options.

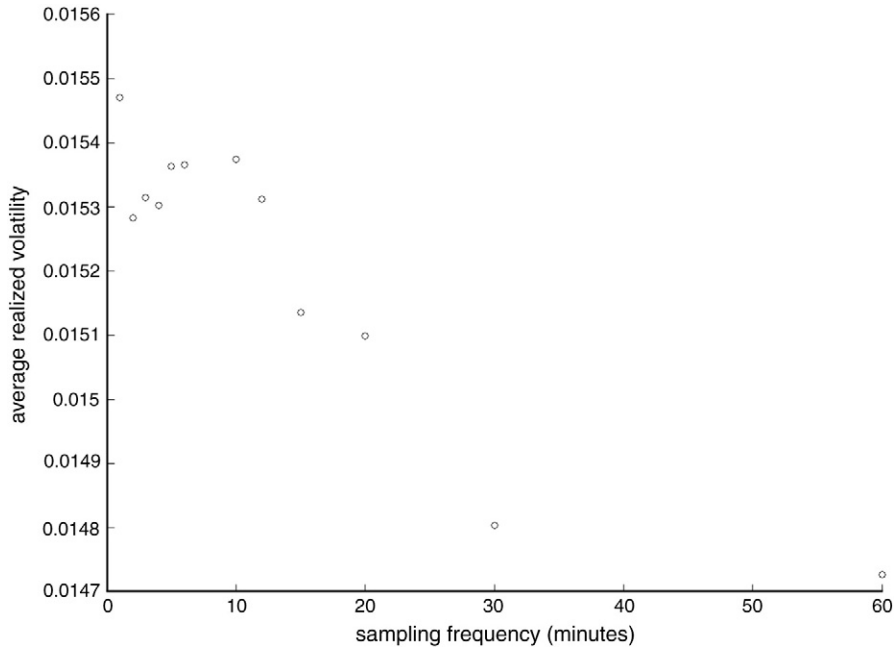

Fig. 1. Volatility signature plot for the period of out-of-sample forecast.

Andersen et al. (2001a) and Andersen et al. (1999), the volatility signature plot is the plot of average realized volatility against sampling frequency m. From Fig. 1, it appears that the average realized volatility for the sampling frequency from 1 min to 12 min contains the microstructure noise. Fig. 2 shows the RMSE (root-mean-square-error) against sampling frequency m for the period of out-of-sample forecast following Aït-Sahalia et al. (2005). We use the model with the theorem 1 from Aït-Sahalia et al. (2005).[7] From Fig. 2, the value of RMSE decreases slightly from the sampling frequency of 15 min. Therefore, we choose the 15-minute returns to calculate the realized volatility. This result is consistent with Chevallier and Sévi (2011). We use the daily transaction record from 07:15 (GMT) to 16:45 (GMT), and the price at the 15-minute mark is obtained by linearly interpolating from the average of the log bid and the log ask for the two closest ticks, following Andersen et al. (2001b). The data for ECX carbon futures daily closing prices, quoted IV, and intraday tick prices are obtained from ECX official website.

The Brent oil variable is the volatility using daily log returns on the Intercontinental Exchange (ICE) Generic 1st Crude Oil Brent 1-month futures[8] in $/barrel. The natural gas variable is the volatility using daily log returns on the ICE UK Natural Gas 1-month futures in /Therm.[9] The Brent oil prices and ICE natural gas futures prices are the reference prices within the European Union for crude oil market products and natural gas market products (Chevallier, 2011a). The coal variable is the volatility using daily log returns on the European Energy Exchange (EEX) Coal ARA 1-month futures in $/ton. The electricity variable is the volatility using daily log returns on the EEX Germany/Austria Phelix 1-month futures in €/MWh. For the consistency, the prices of Brent oil futures, natural gas futures, and coal futures are converted to € using the daily exchange rate from the European Central Bank (ECB).[10] Following Agnolucci (2009), we use absolute values of daily log returns of the Brent oil, natural gas, coal, and electricity variables as volatility values in the same way as the data for the k-NN method. The data sources for the above variables are following: Brent oil, coal, and electricity from the Bloomberg database, and natural gas from the ICE official website. Table 1 shows summary statistics of daily log returns data, and Table 2 shows the matrix of cross-correlations between energy variables. From Table 2, we can find that the values of correlation coefficients are below 20%, and energy variables are not highly correlated. Moreover, following Chevallier (2009),[11] we cannot reveal problematic multi-colinearities between energy variables.

In-sample estimation parameters in Eq. (22) are estimated on a 100-day rolling sample. In other words, we use a window that shifts forward as new information arrives in the same manner as the GARCH-type forecast. The only difference is that a 100-day rolling sample consists of the best forecast among GARCH-type and IV models, Brent oil, natural gas, coal, and electricity volatilities. When estimating in-sample parameters using a 100-day rolling sample from October 09, 2009 to March 02, 2010, we can obtain a one-day-ahead forecast for March 03, 2010 by inserting estimated parameters into Eq. (21). By then shifting the estimation window by one day and re-estimating the model over the new sample, we can obtain a forecast for March 04, 2010. This estimation process ends when we obtain a forecast for August 15, 2011.

4. Results from the GARCH-type models estimation

We estimate GARCH, EGARCH, TGARCH, and GJR-GARCH models with normal and Student's t-distributions for ε_t in Eqs. (1) to (6) with a number of AR, MA, and GARCH terms varying between zero and three; ARCH, and asymmetric terms vary from one to three. Bollerslev and Wooldridge (1992) robust standard errors are used in inferences for estimating GARCH-type models. In the case of the regressions that investigate the information content of forecasts, Newey and West (1987) standard errors are used. Models are selected according to the value of the information criteria. The performance of the models is assessed on the basis of the following diagnostic tests: a Q test on the standardized and squared standardized residuals, an ARCH test, and a Jarque–Bera test. Although the search specification is conducted on

[7] Following Aït-Sahalia et al. (2005), the model with the theorem 1 is used to calculate the RMSE of the realized volatility proxy when the presence of the noise is ignored. For the calculation of the RMSE, we use 1.52% and 0.023% as the value of the true daily volatility and the absolute value of covariance of daily log return, respectively.

[8] Although Chevallier (2011a) uses the price of Crude Oil-Brent Dated Free on Board, for consistency, we employ the Brent oil futures daily log returns as the Brent oil variable in this article.

[9] 1 Therm indicates 29.3071 kWh.

[10] Available at http://www.ecb.int/stats/exchange/eurofxref/html/index.en.html.

[11] Chevallier (2009) documents that if every simple correlation is below 60% in absolute value, the matrix of cross-correlations would not reveal problematic multi-colinearities between variables.

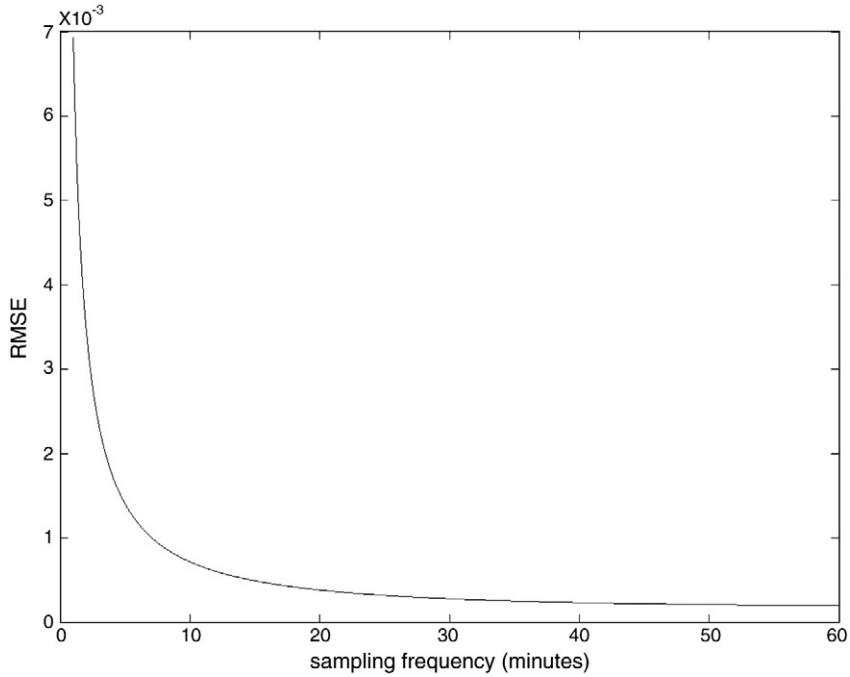

Fig. 2. RMSE against sampling frequency *m* for the period of out-of-sample forecast following Aït-Sahalia et al. (2005).

Table 1
Summary statistics of daily log returns data.

	Mean	Median	Max	Min	Std. Dev.	Skewness	Kurtosis	Obs.
ECX full sample	−0.0007	0.0000	0.1137	−0.1003	0.0239	−0.0866	5.5471	922
ECX for GARCH	−0.0012	−0.0015	0.1137	−0.0943	0.0281	0.0957	4.6259	452
ECX after GARCH	−0.0002	0.0007	0.0614	−0.1003	0.0191	−0.5355	6.1116	470
Brent oil	0.0011	0.0009	0.0580	−0.0680	0.0170	−0.1642	3.8328	470
Coal	0.0012	0.0012	0.0637	−0.0560	0.0151	0.3278	5.4319	470
Natural gas	0.0011	−0.0009	0.1848	−0.0957	0.0280	1.2383	8.7720	470
Electricity	0.0001	−0.0017	0.1716	−0.1446	0.0237	2.2851	23.3272	470

Note: ECX full sample refers to the first log-differenced ECX EUA carbon futures contracts daily closing prices from January 03, 2008 to August 15, 2011, ECX for GARCH refers to the first log-differenced ECX EUA carbon futures contracts daily closing prices from January 03, 2008 to October 08, 2009, ECX after GARCH refers to the first log-differenced ECX EUA carbon futures contracts daily closing prices from October 09, 2009 to August 15, 2011, Brent oil refers to the first log-differenced Generic 1st Crude Oil Brent 1-month futures daily closing prices from October 08, 2009 to August 12, 2011, Coal refers to the first log-differenced coal ARA 1 month futures daily closing prices from October 08, 2009 to August 12, 2011, Natural gas refers to the first log-differenced UK natural gas 1 month futures daily closing prices from October 08, 2009 to August 12, 2011, Electricity refers to the first log-differenced EEX Germany/Austria Phelix electricity 1 month futures daily closing prices from October 08, 2009 to August 12, 2011, Std. Dev. refers to the standard deviation, and Obs. refers to the number of observations.

the basis of the information criteria, the models with the lowest value of the information criteria are rejected when the Q test with 5, 10, or 15 lags for the squared standardized residuals is significant at the 99% confidence level. In this case, the models with the second lowest value of the information criteria are examined. This selection process proceeds to the model with the third lowest value of the information criteria. When none of these models passes the Q test, no model is selected. In other words, a maximum of six models is examined before rejecting a model, i.e., three models selected by the AIC and three by the BIC.

Tables 3, 4, 5, and 6 show the estimation results of GARCH, EGARCH, TGARCH, and GJR-GARCH models, respectively. In Tables 3, 4, and 6, Normal1 and Normal2 indicate the best models among those models assumed to have normal residuals according to AIC and BIC, respectively. Similarly, T1 and T2 indicate the best models of those assumed to have Student's *t* residuals according to AIC and BIC, respectively. First of all, with respect to the values of information criteria, more complicated GARCH, TGARCH, and GJR-GARCH specifications do not seem to improve on the widely used GARCH(1,1), TGARCH(1,1), and GJR-GARCH(1,1). With respect to the EGARCH models, the most complicated EGARCH models, i.e., EGARCH(3,3), are selected, and most of the estimated parameters for the conditional variance equation are statistically significant. Except for the case of EGARCH specifications, this result is consistent with the empirical results for carbon spot prices of Benz and Trück (2009).[12] In other words, the mean equation error term from the return of carbon futures prices does not have constant variance, but the conditional variance. Therefore, the volatility of the carbon futures return is affected by the previous value, and has a persistent characteristic, which means that the volatility of carbon futures return is easy to forecast the future value. Secondly, for GARCH, TGARCH, and GJR-GARCH specifications, the estimated parameters for AR and MA terms, which have one or two terms, are significant at the 5% significance level. This result indicates that the return of carbon futures is affected by the two previous values and the two previous shocks to the return, and has a persistent characteristic similar to the volatility. However, in case of three AR and MA terms, the estimated parameters for the first AR and MA terms are insignificant, and those for the second and third AR and MA terms are significant at the 5% level. This result seems to be an over-parameterization phenomenon of models.

[12] Benz and Trück (2009) find that, for AR, the estimated parameters of higher orders have limited explanatory power. The authors also present that the estimated parameters of higher orders for GARCH are not significant.

Table 2
Matrix of cross-correlations between energy variables.

	Brent oil	Coal	Natural gas	Electricity
Brent oil	1			
Coal	0.0686	1		
Natural gas	0.0243	0.1732	1	
Electricity	0.1098	0.1745	0.1996	1

Note: Brent oil refers to the first log-differenced Generic 1st Crude Oil Brent 1-month futures daily closing prices from October 08, 2009 to August 12, 2011, Coal refers to the first log-differenced coal ARA 1 month futures daily closing prices from October 08, 2009 to August 12, 2011, Natural gas refers to the first log-differenced UK natural gas 1 month futures daily closing prices from October 08, 2009 to August 12, 2011, Electricity refers to the first log-differenced EEX Germany/Austria Phelix electricity 1 month futures daily closing prices from October 08, 2009 to August 12, 2011.

In contrast, for EGARCH models, the estimated parameters for MA(2) are statistically insignificant, but those for ARMA(3,1) are significant at the 1% level. The characteristics of estimated parameters in EGARCH models are contrary to those in GARCH, TGARCH, and GJR-GARCH models. Thirdly, the value of kurtosis reduces through the estimation of GARCH-type models. See the values of the kurtosis in Tables 1, 3, 4, 5, and 6. This result shows that the value of kurtosis approaches 3, which is the kurtosis of a normal distribution. Therefore, after the estimation of parameters, the distribution of the residuals approaches a white noise. Whether the residuals are assumed to be a normal or Student's *t*-distribution does not significantly affect the value of kurtosis. Finally, except for the EGARCH(3,3) model, the choice of distribution type for the residuals does not affect the estimated parameters for conditional variance equation. In other words, the estimated parameters for the conditional variance equation of each GARCH-type model vary little along the distribution type of the residuals.

Table 3
Estimation results of the GARCH models.

	Normal1	Normal2	T1	T2
Mean equation				
Constant in mean	0.0034	0.0003	0.0010	0.0007
	(0.4703)	(0.1387)	(0.4512)	(0.4202)
AR(1)	0.1123	−0.4739***	0.1902*	−0.5025***
	(0.7248)	(−4.7844)	(1.7256)	(−4.9176)
AR(2)	−0.7123***		−0.6282***	
	(−12.0547)		(−8.9946)	
AR(3)	−0.4457**		−0.4316***	
	(−2.4059)		(−3.6720)	
MA(1)	0.0297	0.6395***	−0.0446	0.6627***
	(0.1972)	(7.9541)	(−0.5211)	(8.1461)
MA(2)	0.6075***		0.5386***	
	(11.1216)		(9.4405)	
MA(3)	0.6304***		0.6409***	
	(5.2116)		(7.2463)	
Conditional variance equation				
ω	3.345×10^{-5}	2.9799×10^{-5}	2.3904×10^{-5}	2.5655×10^{-5}
	(0.8336)	(0.9876)	(1.0894)	(1.2841)
α_1	0.1360	0.1069	0.1390*	0.1349**
	(1.0880)	(1.4903)	(1.8561)	(2.1395)
β_1	0.8208***	0.8533***	0.838***	0.8384***
	(4.9915)	(8.1325)	(9.2481)	(10.9508)
Distribution parameter			7.648***	8.4916***
			(2.7220)	(2.9026)
Diagnostics				
ARCH test	0.8179	0.8618	0.9666	0.7722
Skewness	−0.2818	−0.3305	−0.3856	−0.3624
Kurtosis	3.5635	3.7246	3.9073	3.8658
JB test	11.9604***	18.1158***	26.7047***	24.0096***
AIC	−2031.7	−2026.9	−2038.1	−2034.0
BIC	−1990.5	−2002.3	−1992.9	−2005.2
Correlation of residuals				
Q(5)	0.8789	0.8150	0.9243	0.7927
Q(12)	0.9235	0.9032	0.9242	0.9054
Q(17)	0.9343	0.7738	0.9286	0.8111
Correlation of squared residuals				
Q(5)	0.8873	0.5798	0.6927	0.5416
Q(12)	0.7624	0.4997	0.8301	0.5368
Q(17)	0.6263	0.3964	0.6461	0.4943

Note: *** indicates significance at 1%, ** at 5%, and * at 10%. Normal1 and Normal2 indicate the best models in the models assumed to have normal residuals according to AIC and BIC, respectively. T1 and T2 indicate the best models in the models assumed to have Student's *t* residuals according to AIC and BIC, respectively. The *t*-statistics are in parentheses below corresponding parameter estimates. In the case of ARCH test and correlation of residuals and squared residuals, the table reports the *p*-value for the null hypothesis. Other statistics including Jarque–Bera (JB) test indicate the values of the statistics.

Table 4
Estimation results of the EGARCH models.

	Normal1	Normal2	T1	T2
Mean equation				
Constant in mean	−0.0007	−0.0008	−0.0006	−0.0005
	(−0.1987)	(−0.7152)	(−1.0279)	(−0.5057)
AR(1)			−0.2587***	
			(−27.2995)	
AR(2)			−0.0690***	
			(−16.0518)	
AR(3)			0.0511***	
			(10.3497)	
MA(1)	0.1422		0.3899***	
	(1.0484)		(19.1969)	
MA(2)	−0.0934			
	(−0.2170)			
Conditional variance equation				
ω	−0.5021	−0.3630*	−1.1359***	−0.3356*
	(−1.5486)	(−1.7157)	(−242.4848)	(−1.7535)
α_1	−0.0924	−0.1322***	−0.1187***	−0.1316***
	(−0.7813)	(−3.3901)	(−58.7310)	(−3.3765)
α_2	0.1081		−0.1487***	
	(1.0016)		(−10.2662)	
α_3	−0.1241		−0.1698***	
	(−0.9031)		(−11.1451)	
β_1	2.0953***	0.9505***	−0.5098***	0.9547***
	(36.2300)	(32.7731)	(−127.1963)	(36.4931)
β_2	−2.0386***		0.4566***	
	(−154.3620)		(−95.8394)	
β_3	0.8767***		0.9041***	
	(38.2104)		(4846.0986)	
γ_1	0.1581	0.1899*	0.2826***	0.2005**
	(1.4059)	(1.8263)	(9.1649)	(2.2165)
γ_2	−0.0535		0.2517***	
	(−0.1647)		(14.4882)	
γ_3	0.1160		−0.1004***	
	(0.4142)		(−18.5777)	
Distribution parameter			19.275***	13.47**
			(12.0390)	(1.9763)
Diagnostics				
ARCH test	0.9356	0.7129	0.5912	0.8184
Skewness	−0.1630	−0.1914	−0.2130	−0.2041
Kurtosis	3.0832	3.3727	3.2629	3.4092
JB test	2.1320	5.3752*	4.7209*	6.2923**
AIC	−2065.0	−2037.3	−2059.5	−2038.2
BIC	−2011.5	−2016.7	−1993.7	−2013.5
Correlation of residuals				
Q(5)	0.7223	0.0295	0.9226	0.0319
Q(12)	0.8695	0.1556	0.8724	0.1681
Q(17)	0.7792	0.1411	0.6744	0.1641
Correlation of squared residuals				
Q(5)	0.6918	0.3576	0.5065	0.3392
Q(12)	0.8399	0.3868	0.7074	0.3840
Q(17)	0.8636	0.5114	0.8455	0.5089

Note: *** indicates significance at 1%, ** at 5%, and * at 10%. Normal1 and Normal2 indicate the best models in the models assumed to have normal residuals according to AIC and BIC, respectively. T1 and T2 indicate the best models in the models assumed to have Student's *t* residuals according to AIC and BIC, respectively. The *t*-statistics are in parentheses below corresponding parameter estimates. In the case of ARCH test and correlation of residuals and squared residuals, the table reports the *p*-value for the null hypothesis. Other statistics including Jarque–Bera (JB) test indicate the values of the statistics.

Table 5
Estimation results of the TGARCH models.

	Normal	T
Mean equation		
Constant in mean	−0.0008	−0.0006
	(−0.5195)	(−0.5003)
MA(1)	0.1201**	0.1175**
	(2.4361)	(2.3367)
MA(2)	−0.1270**	−0.1284**
	(−2.0189)	(−2.3937)
Conditional variance equation		
ω	0.0019	0.0019
	(0.3246)	(0.5555)
α_1^+	1.1399×10^{-21}	0
	(0.0000)	(0.0000)
α_1^-	0.1260	0.1281
	(0.8555)	(1.4927)
β_1	0.8740***	0.8719***
	(2.7902)	(4.6551)
Distribution parameter		17.304
		(1.4298)
Diagnostics		
ARCH test	0.1416	0.1426
Skewness	−0.1466	−0.1428
Kurtosis	3.2705	3.2729
JB test	2.9970	2.9381
AIC	−2039.3	−2039.2
BIC	−2010.5	−2006.3
Correlation of residuals		
Q(5)	0.7919	0.7840
Q(12)	0.8827	0.8819
Q(17)	0.5834	0.5817
Correlation of squared residuals		
Q(5)	0.1098	0.1077
Q(12)	0.1274	0.1311
Q(17)	0.0978	0.1000

Note: *** indicates significance at 1%, and ** at 5%. Normal1 and Normal2 indicate the best models in the models assumed to have normal residuals according to AIC and BIC, respectively. T1 and T2 indicate the best models in the models assumed to have Student's *t* residuals according to AIC and BIC, respectively. The *t*-statistics are in parentheses below corresponding parameter estimates. In the case of ARCH test and correlation of residuals and squared residuals, the table reports the *p*-value for the null hypothesis. Other statistics including Jarque–Bera (JB) test indicate the values of the statistics.

Table 6
Estimation results of GJR-GARCH models.

	Normal1	Normal2	T1	T2
Mean equation				
Constant in mean	0.0011	−0.0015	−0.0010	−0.0009
	(0.2889)	(−0.7659)	(−0.3615)	(−0.5287)
AR(1)	0.1009	−0.4931***	0.1939	−0.5109***
	(0.9649)	(−5.2771)	(0.9310)	(−5.4483)
AR(2)	−0.7002***		−0.6276***	
	(−11.0653)		(−4.1867)	
AR(3)	−0.4753***		−0.4329**	
	(−4.1260)		(−2.1922)	
MA(1)	0.0254	0.6463***	−0.0473	0.6653***
	(0.3079)	(8.4623)	(−0.2728)	(8.7905)
MA(2)	0.6025***		0.5401***	
	(12.4035)		(4.4870)	
MA(3)	0.6433***		0.6391***	
	(6.5949)		(3.7978)	
Conditional variance equation				
ω	1.6253×10^{-5}	2.133×10^{-5}	1.9594×10^{-5}	$1.9929 \times 10^{-5*}$
	(1.1297)	(1.1427)	(0.7581)	(1.7099)
α_1	0	0	0	0
	(0.0000)	(0.0000)	(0.0000)	(0.0000)
β_1	0.8950***	0.8869***	0.8772***	0.8814***
	(11.4094)	(9.5194)	(4.5735)	(15.6258)
γ_1	0.1508***	0.1646***	0.1922***	0.1830***
	(2.6174)	(2.5821)	(2.5896)	(2.9253)
Distribution parameter			11.17*	12.518**
			(1.8566)	(2.0706)
Diagnostics				
ARCH test	0.6651	0.9574	0.9753	0.8663
Skewness	−0.1950	−0.2402	−0.2778	−0.2511
Kurtosis	3.2280	3.4433	3.5707	3.5211
JB test	3.8432	8.0467**	11.9477***	9.8638**
AIC	−2050.0	−2046.4	−2052.4	−2048.2
BIC	−2004.7	−2017.7	−2003.1	−2015.2
Correlation of residuals				
Q(5)	0.9881	0.8845	0.9424	0.9008
Q(12)	0.9877	0.9314	0.9434	0.9381
Q(17)	0.9744	0.7756	0.9474	0.7960
Correlation of squared residuals				
Q(5)	0.6172	0.4806	0.5040	0.4323
Q(12)	0.5790	0.5542	0.7033	0.5225
Q(17)	0.6925	0.6577	0.6999	0.6386

Note: *** indicates significance at 1%, ** at 5%, and * at 10%. Normal1 and Normal2 indicate the best models in the models assumed to have normal residuals according to AIC and BIC, respectively. T1 and T2 indicate the best models in the models assumed to have Student's *t* residuals according to AIC and BIC, respectively. The *t*-statistics are in parentheses below corresponding parameter estimates. In the case of ARCH test and correlation of residuals and squared residuals, the table reports the *p*-value for the null hypothesis. Other statistics including Jarque–Bera (JB) test indicate the values of the statistics.

The estimated parameters for the conditional variance equation in Tables 4 and 6 indicate that the asymmetric terms are statistically significant. Except for the EGARCH-Normal1 and TGARCH models, the asymmetric terms are significant at the 10% level. This result indicates that a negative shock to the return of carbon futures leads to a higher conditional variance of the return in the subsequent period than a positive shock, and confirms Chevallier (2009)[13] and Chevallier (2011b).[14] Considering the *p*-values in Tables 3, 4, 5, and 6, the use of GARCH-type models does not result in serial correlation in the standardized residuals and squared standardized residuals. Indeed, the *p*-values for GJR-GARCH model are greater than those for the EGARCH and TGARCH models. Therefore, from all results, we can conclude that the GJR-GARCH model performs better than the EGARCH and TGARCH models.

5. Results from the model comparison

Tables 7 to 10 report the results from the model comparison for the forecasts. Table 7 shows the loss function values, such as MSE, MSE-LOG, MAE, MAE-LOG, and QLIKE, for the GARCH-type, IV, and k-NN models. According to the loss function values, GARCH-type models perform better than IV and k-NN models. The overall ranks of IV for all loss functions are 15 or 16, i.e., forecasts using IV have lower predictive accuracy than GARCH-type models. Similarly, except for the value of QLIKE, k-NN model also has lower predictive accuracy than GARCH-type models. When we add all rank values of loss functions for each model, three lowest values are found in the models of GJR-Normal1, GARCH-Normal1, and GJR-Normal2. Therefore, these three models have better predictability about the carbon futures volatility than other models.

Table 8 reports the estimation results of Eq. (14). Here, we can examine the information content of the forecasts using the Mincer–Zarnowitz regression. According to the values of adjusted R^2, AIC, and BIC, GJR-Normal1 has the lowest AIC and BIC values, and the highest adjusted R^2 values. Therefore, GJR-Normal1 is the best model. On the other hand, IV and k-NN models have the lowest values of the adjusted R^2 and the highest values of the AIC and BIC. Therefore, IV and k-NN models have the poorest information content about the realized carbon futures volatility among all models, which is consistent with the results shown

[13] Following Chevallier (2009), the terms for asymmetric effect are significant at the 1% level.

[14] Chevallier (2011b) finds that the asymmetric effect is statistically significant through the likelihood ratio (LR) test.

Table 7
MSE, MSE-LOG, MAE, MAE-LOG and QLIKE from the GARCH, EGARCH, TGARCH, GJR-GARCH, IV, and k-NN models.

	MSE		MSE-LOG		MAE		MAE-LOG		QLIKE	
	Value	Overall rank	Value	Overall rank	Value	Overall rank	Value	Overall rank	Value	Overall rank
GARCH										
Normal1	5.635×10^{-5}	2	0.1814	2	5.614×10^{-3}	2	0.3462	3	−2.896	10
Normal2	5.648×10^{-5}	4	0.1836	7	5.673×10^{-3}	6	0.3512	8	−2.889	12
T1	5.830×10^{-5}	6	0.1821	5	5.651×10^{-3}	5	0.3464	4	−2.901	6
T2	5.839×10^{-5}	7	0.1837	8	5.686×10^{-3}	8	0.3497	7	−2.894	11
EGARCH										
Normal1	7.354×10^{-5}	13	0.2160	12	6.234×10^{-3}	12	0.3788	12	−2.866	13
Normal2	5.682×10^{-5}	5	0.1888	10	5.722×10^{-3}	9	0.3574	10	−2.896	9
T1	7.458×10^{-5}	14	0.2087	11	6.122×10^{-3}	11	0.3657	11	−2.906	4
T2	5.873×10^{-5}	9	0.1872	9	5.753×10^{-3}	10	0.3558	9	−2.899	8
TGARCH										
Normal	6.711×10^{-5}	11	0.2472	14	6.690×10^{-3}	13	0.4227	14	−2.778	15
T	6.918×10^{-5}	12	0.2468	13	6.706×10^{-3}	14	0.4226	13	−2.783	14
GJR-GARCH										
Normal1	5.446×10^{-5}	1	0.1769	1	5.537×10^{-3}	1	0.3430	1	−2.910	3
Normal2	5.636×10^{-5}	3	0.1818	4	5.635×10^{-3}	4	0.3487	6	−2.900	7
T1	5.887×10^{-5}	10	0.1827	6	5.623×10^{-3}	3	0.3439	2	−2.912	2
T2	5.865×10^{-5}	8	0.1816	3	5.677×10^{-3}	7	0.3476	5	−2.905	5
IV	2.344×10^{-4}	15	3.8886	16	1.275×10^{-2}	15	1.8773	16	−4.109	16
k-NN	6.333×10^{-4}	16	2.4117	15	1.417×10^{-2}	16	1.0321	15	−3.040	1

Note: For each criterion, the first column shows the value of the criterion, and the second column shows the rank of the models across types including IV and k-NN model. The QLIKE value of the realized volatility is −3.296. Therefore, the rank in the QLIKE column is defined by how the QLIKE value of the forecasting model is different from that of the realized volatility.

in Table 7. The IV result seems to be due to the low trading volume of carbon options.[15] Poor forecasting performance of the k-NN model probably indicates that the process of carbon futures volatility does not have same pattern as the past movement, and is affected by some shocks from the exterior condition. Most estimated values for parameter α are insignificant, however, all estimated values for parameter β, with the exception of the IV model, are significant at the 1% level. This result indicates that all GARCH-type models have good predictability about the volatility of carbon futures. However, the value of β is not the same as the theoretical value of 1. Moreover, as shown by the values of the F test, all estimators are biased. From these results, we can find that there is a room for the improvement of forecasting carbon futures volatility. This improvement is carried out by using energy volatilities in the next section. The values for the adjusted R^2 are noticeably higher than those of Chevallier (2010) and Chevallier and Sévi (2011).[16]

With respect to the results of Tables 7 and 8, the relative efficiency of the best three GARCH-type models, i.e., GJR-Normal1, GJR-Normal2, and GARCH-Normal1, IV, and k-NN model is presented in Table 9. The estimated parameters for IV are insignificant, confirming that IV is not an efficient forecaster for the volatility of carbon futures. In contrast, the estimated parameters for k-NN model are statistically significant. However, when explanatory variables consist of k-NN model and the best three GARCH-type models, significant improvement in the information content is not found, compared to the result of GJR-Normal1 in Table 8. Except for the case of k-NN model, all estimated parameters for α and β_2 are insignificant. Therefore, the combined forecasting models do not have the superiority in the predictability about carbon futures volatility, compared to other single forecasting models. However, for β_1, several parameters are statistically significant. According to the values of adjusted R^2, when the regression equation contains GJR-Normal1 as an explanatory variable, the values of adjusted R^2 are higher than other regression cases. Similarly, the values of AIC and BIC with GJR-Normal1 are lower than other cases. Moreover, the values of adjusted R^2, AIC, and BIC with GJR-Normal1 in Table 9 are not significantly different from those values for GJR-Normal1 in Table 8. From these results, we can conclude that GJR-Normal1 outperforms all models, including IV, k-NN, and GARCH-type models. In other words, the GJR-Normal1 model has the most information content for the volatility of carbon futures.

In Table 10, the results of the Diebold–Mariano test, the superior predictive ability test, and the model confidence set test are shown. From the results of Tables 7, 8, and 9, we define the benchmark model as GJR-Normal1, the best forecast model. According to the negative and statistically significant values of the Diebold–Mariano test statistics of EGARCH, TGARCH, IV, and k-NN models, these models have significantly poorer forecasting abilities than GJR-Normal1. Moreover, for GARCH and GJR models, most of the values of the Diebold–Mariano test statistics are negative, and several values are statistically significant. These results show that GJR-Normal1 is the best model among GARCH-type, IV, and k-NN models in forecasting carbon futures volatility, and this is consistent with the previous results. In the case of the superior predictive ability test, the p-values in Table 10 are more than 0.9, so the null hypothesis, that GJR-Normal1 has the superiority in the predictability about carbon futures volatility, is accepted for every loss function. The results of the model confidence set test indicate that the worst forecasting models for carbon futures volatility is IV model. All models except for IV model have p-values nearly 1, and are included in $\widehat{M}^*_{75\%}$. As a result, GJR-Normal1 is included in $\widehat{M}^*_{75\%}$, and has good forecast ability for the volatility of carbon futures. Fig. 3 shows the comparison of the forecasted value from GJR-Normal1 and the realized volatility of carbon futures.

6. Results from GARCH-type model with energy volatilities

Table 11 indicates the model specification for the forecast with energy variables. The best of the GARCH-type models, GJR-Normal1, is fixed as an explanatory variable. Other energy variables, Brent oil, coal, natural gas, and electricity, are included as explanatory variables. Along with the model specification in Table 11, MSE, MSE-LOG, MAE, MAE-LOG, and QLIKE values of the models are shown in Table 12. The reference model

[15] ECX reports the trading volume of all derivatives on its official website. According to the statistics on the website, we find that ECX EUA options have a much lower trading volume than do EUA futures.

[16] Chevallier (2010) reports 0.144 and Chevallier and Sévi (2011) reports 0.0703 as the value of adjusted R^2, respectively.

Table 8
Estimation results of Eq. (14), i.e., estimated parameters, t-statistics, information criteria, adjusted R^2, and F test.

	α	t_α	β	t_β	AIC	BIC	Adj-R^2	F test
GARCH								
Normal1	−0.0009	−0.5403	0.8396***	8.1332	−10.1400	−10.1223	0.4445	0
Normal2	−0.0012	−0.6813	0.8504***	8.1116	−10.1481	−10.1304	0.4490	0
T1	−0.0003	−0.1752	0.8089***	7.7771	−10.0940	−10.0764	0.4184	0
T2	−0.0005	−0.3073	0.8185***	7.6998	−10.1018	−10.0842	0.4229	0
EGARCH								
Normal1	0.0035***	2.7291	0.6139***	8.3340	−9.8777	−9.8600	0.2780	0
Normal2	−0.0014	−0.6634	0.8754***	6.8978	−10.0730	−10.0553	0.4061	0
T1	0.0041***	3.2924	0.5848***	7.8461	−9.8904	−9.8728	0.2871	0
T2	−0.0005	−0.2772	0.8240***	7.6630	−10.0538	−10.0362	0.3946	0
TGARCH								
Normal	−0.0077**	−2.5417	1.1407***	6.9462	−10.0454	−10.0277	0.3894	0
T	−0.0073**	−2.3747	1.1294***	6.6339	−9.9735	−9.9558	0.3439	0
GJR-GARCH								
Normal1	−0.0002	−0.1266	0.8110***	7.5811	−10.1657	−10.1480	0.4586	0
Normal2	−0.0001	−0.0594	0.7997***	7.2437	−10.1471	−10.1294	0.4485	0
T1	0.0008	0.4105	0.7607***	6.9409	−10.0766	−10.0590	0.4082	0
T2	0.0005	0.2924	0.7662***	6.9046	−10.1051	−10.0875	0.4248	0
IV	0.0145***	13.4378	0.2764	1.2236	−9.5531	−9.5354	0.0011	0
k-NN	0.0151***	16.2098	0.0049	0.2637	−9.5501	−9.5324	−0.0019	0

Note: *** indicates significance at 1%, and ** at 5%. The column for F test indicates p-values for the unbiasedness of each model.

in Table 12 indicates the forecast model that has only GJR-Normal1 as an explanatory variable. The sample period of the reference model is identical to the estimation period of model 1 to 15. Overall, most of the values of MSE, MSE-LOG, MAE, MAE-LOG, and QLIKE are lower than those of the reference model. According to the values of the loss functions, the best models are different depending on the criterion. To determine the best model, we add all rank values of loss functions for each model and find the model with the minimum value. Using this rationale, the best model is determined as model 12, which has Brent oil, coal, and natural gas as explanatory variables. This result shows that the forecast model performs better when the energy variables are included in the model. In other words, the volatilities of energy variables have the volatility spillovers to carbon futures,[17] and the volatilities of other energy markets affect the volatility of carbon futures. As a result, carbon futures market is affected by other energy markets.

The estimation results of Eq. (20) with energy variables are presented in Table 13. According to Eqs. (20) and (21), the realized volatility as a dependent variable is a variable at t, the best GARCH-type forecast for time t using GJR-Normal1 is a variable at $t-1$, and the volatilities of Brent oil, coal, natural gas, and electricity are variables at $t-1$. In Table 13, the estimated parameters for forecasting using GJR-Normal1 and energy variables in all models are significant at the 1% level. Moreover, most of the values of AIC and BIC are lower than those of the reference model, and most of the adjusted R^2 are higher than the reference model. In the case of model 6, the value of adjusted R^2 increases from 0.5552 to 0.5878, a 5.8718% increase. This improvement comes from the inclusion of other energy variables. With regard to the values of the F test, estimators in all models are biased. From the results of Table 13, model 6 has the highest value of adjusted R^2. However, when we add the rank of the adjusted R^2 to the rank values in Table 12, model 12 has the lowest value. Therefore, model 12 is the best forecast model of carbon futures volatility, and the volatilities of Brent oil, coal, and electricity seem to have a better ability to forecast the following day's carbon futures volatility than do other combinations of energy variables. This result confirms that the volatilities of energy markets have the volatility spillovers to carbon futures. Models that have low values of the sum of all ranks, i.e., models 5, 7, 12, and 13, include Brent oil as an explanatory variable.[18] According to this result, Brent oil has more information content about the following day's carbon futures volatility than other energy variables, and presents good forecasting performance. This result is consistent with Chevallier (2011a). The superior forecasting performance of Brent oil can be explained by using the trading volume. According to the Bloomberg database, the trading volume of Brent oil is much higher than those of other energy variables. Therefore, Brent oil has the most information content about energy markets among energy variables, and

Table 9
Estimation results of Eq. (15), i.e., estimated parameters, t-statistics, information criteria, and adjusted R^2, when the proxy for volatility is regressed on an intercept and two forecasts obtained from the IV model, k-NN model, and the four GARCH-type models.

	α	β_1	β_2	AIC	BIC	Adj-R^2
GJR1 vs. IV	−0.0002	0.8115***	−0.0191	−10.1615	−10.1350	0.4575
	(−0.1120)	(7.4648)	(−0.1156)			
GJR2 vs. IV	−0.0001	0.7998***	0.0073	−10.1406	−10.1141	0.4461
	(−0.0822)	(7.1285)	(0.0450)			
GARCH1 vs. IV	−0.0006	0.8320***	−0.0464	−10.1330	−10.1065	0.4418
	(−0.3982)	(7.9441)	(−0.2864)			
GJR1 vs. k-NN	−0.0000	0.8251***	−0.0295***	−10.1742	−10.1477	0.4643
	(−0.0671)	(12.1100)	(−30.8820)			
GJR2 vs. k-NN	0.0001	0.8162***	−0.0317*	−10.1550	−10.1285	0.4540
	(0.0463)	(7.1651)	(−1.7852)			
GARCH1 vs. k-NN	−0.0006	0.8544***	−0.0378***	−10.1530	−10.1265	0.4529
	(−0.3126)	(5.9581)	(10.3796)			
GJR1 vs. GARCH1	−0.0005	0.6401	0.1837	−10.1638	−10.1373	0.4588
	(−0.3412)	(1.4366)	(0.4740)			
GJR1 vs. GJR2	−0.0001	1.3329**	−0.5241	−10.1653	−10.1388	0.4596
	(−0.0635)	(2.0025)	(−0.8047)			
GJR2 vs. GARCH1	−0.0007	0.4574	0.3716	−10.1527	−10.1262	0.4527
	(−0.4318)	(1.3220)	(1.3440)			

Note: *** indicates significance at 1%, ** at 5%, and * at 10%. The t-statistics are in parentheses below corresponding parameter estimates. GJR1, GJR2, and GARCH1 refer to GJR-Normal1, GJR-Normal2, and GARCH-Normal1, respectively. β_1 indicates the estimated parameter for the former in the first column, and β_2 indicates the estimated parameter for the latter in the first column.

[17] In contrast, according to Oberndorfer (2009), the volatility of carbon spot has no volatility spillover effect to the electricity stock portfolio.

[18] Although we do not report the results, we also use the daily log returns of Crude Oil-Brent Dated Free on Board as the Brent oil variable for this study. However, the results are similar to the results from the daily log returns of Brent oil futures. Low values of the sum of all ranks are found in models 5, 7, 12, and 13.

Table 10
Diebold–Mariano test statistics, superior predictive ability test *p*-values, and model confidence set *p*-values for the GARCH, EGARCH, TGARCH and GJR-GARCH, IV, and k-NN models.

	MSE		MSE-LOG		MAE		MAE-LOG		QLIKE	
GARCH										
Normal1	−0.782	1.000**	−1.071	1.000**	−0.734	1.000**	−0.661	1.000**	−0.850	1.000**
Normal2	−0.891	1.000**	−1.807*	1.000**	−1.363	1.000**	−1.818*	1.000**	−1.908*	1.000**
T1	−1.264	1.000**	−1.261	1.000**	−1.057	1.000**	−0.703	1.000**	−0.714	1.000**
T2	−1.401	1.000**	−1.761*	1.000**	−1.485	1.000**	−1.438	1.000**	−1.451	1.000**
EGARCH										
Normal1	−2.565**	1.000**	−4.636***	1.000**	−3.971***	1.000**	−4.131***	1.000**	−4.029***	1.000**
Normal2	−0.666	1.000**	−2.826***	1.000**	−1.358	1.000**	−2.858***	1.000**	−2.446**	1.000**
T1	−2.657***	1.000**	−3.955***	1.000**	−3.165***	1.000**	−2.460**	1.000**	−2.672***	1.000**
T2	−1.100	1.000**	−2.588***	1.000**	−1.923*	1.000**	−2.773***	1.000**	−2.263**	1.000**
TGARCH										
Normal	−2.354**	1.000**	−6.306***	1.000**	−4.612***	1.000**	−6.611***	1.000**	−6.394***	0.999**
T	−2.261**	1.000**	−6.367***	1.000**	−4.786***	1.000**	−6.786***	1.000**	−6.406***	1.000**
GJR-GARCH										
Normal1	–	1.000**	-	1.000**	–	1.000**	–	1.000**	–	1.000**
Normal2	−1.957*	1.000**	−2.878***	1.000**	−2.290**	1.000**	−2.322**	1.000**	−2.690***	1.000**
T1	−1.672*	1.000**	−0.538	1.000**	−0.892	1.000**	−0.019	1.000**	0.162	1.000**
T2	−2.477**	1.000**	−1.897*	1.000**	−2.244**	1.000**	−1.466	1.000**	−1.457	1.000**
IV	−5.465***	0.359**	−17.077***	0.069	−11.001***	0.532**	−23.320***	0.030	−11.521***	0.703**
k-NN	−3.325***	1.000**	−8.716***	1.000**	−7.729***	1.000**	−11.593***	1.000**	−7.397***	1.000**
SPA_u	0.984		0.997		0.997		0.996		0.994	
SPA_c	0.975		0.963		0.975		0.965		0.969	
SPA_l	0.911		0.928		0.947		0.955		0.947	

Note: For each criterion, the first column shows the value of the Diebold–Mariano test statistics, and the second column shows the model confidence set *p*-values. *** indicates significance at 1%, ** at 5%, and * at 10% in the first column. In the second column, ** indicates that the forecast model is included in $\widehat{M}^*_{75\%}$, and * indicates that the forecast model is included in $\widehat{M}^*_{90\%}$. The null hypotheses of these Diebold–Mariano statistics are that each model and GJR-Normal1 have equal predictive accuracy. The benchmark model for the superior predictive ability test is GJR-Normal1.

also has the most information content about the carbon futures market among energy variables.

The results of the Diebold–Mariano test, the superior predictive ability test, and the model confidence set test are shown in Table 14. The benchmark model is model 12, the best performance model. Most of the values of the Diebold–Mariano test statistics are negative, but statistically insignificant. Several test statistics are significant at 10%. This result shows that the improvement through introducing model 12 is not statistically significant. However, for the reference model, the Diebold–Mariano test statistics are clearly negative. Therefore, model 12 has marginal superiority in the predictability about carbon futures volatility. In the case of the superior predictive ability test, all *p*-values in Table 14 are more than 0.48. These high *p*-values indicate that the null hypothesis, that model 12 outperforms other models in the predictability about carbon futures volatility, is accepted for every loss function. The results of the model confidence set test indicate that all models have *p*-values nearly 1, and are included in $\widehat{M}^*_{75\%}$. In other words, the predictability of models 1 to 15 and the reference model is not significantly different each other. However, because model 12 is included in $\widehat{M}^*_{75\%}$, and has good forecast ability for the volatility of carbon futures according to the results of Tables 12 and 13, we can

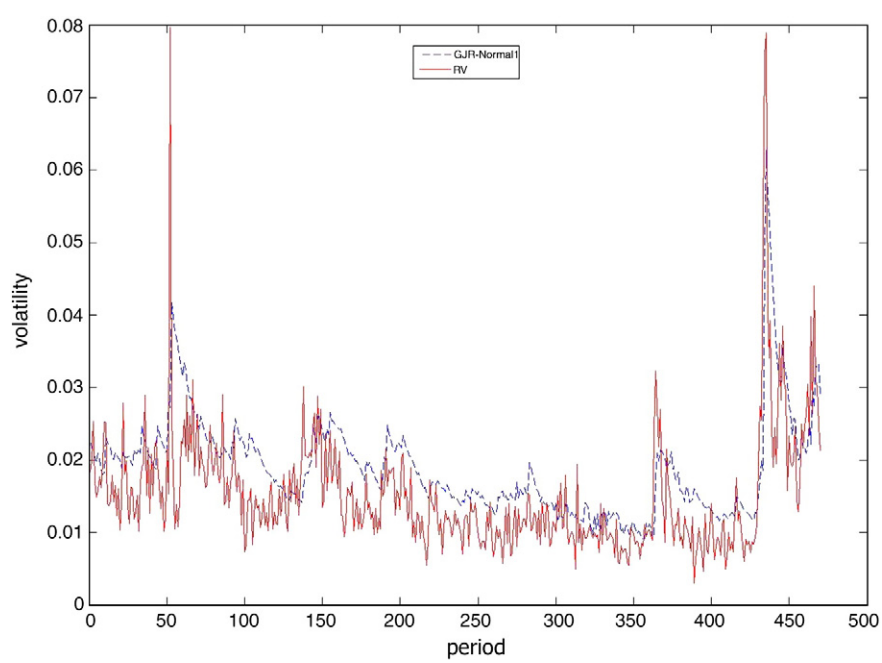

Fig. 3. The forecasted value from GJR-Normal1, the best forecast model from GARCH-type models, and the realized volatility of carbon futures.

Table 11
Model specifications for the forecast with energy variables.

	Explanatory variables				
Model 1	GJR-Normal1	Brent oil			
Model 2	GJR-Normal1		Coal		
Model 3	GJR-Normal1			Natural gas	
Model 4	GJR-Normal1				Electricity
Model 5	GJR-Normal1	Brent oil	Coal		
Model 6	GJR-Normal1	Brent oil		Natural gas	
Model 7	GJR-Normal1	Brent oil			Electricity
Model 8	GJR-Normal1		Coal	Natural gas	
Model 9	GJR-Normal1		Coal		Electricity
Model 10	GJR-Normal1			Natural gas	Electricity
Model 11	GJR-Normal1	Brent oil	Coal	Natural gas	
Model 12	GJR-Normal1	Brent oil	Coal		Electricity
Model 13	GJR-Normal1	Brent oil		Natural gas	Electricity
Model 14	GJR-Normal1		Coal	Natural gas	Electricity
Model 15	GJR-Normal1	Brent oil	Coal	Natural gas	Electricity

Note: the GJR-Normal1 refers to the best forecast from the GARCH-type estimation. Brent oil, coal, natural gas, and electricity indicate the realized volatilities of the previous day, respectively.

conclude that the best forecast model is model 12. When we forecast the future volatility of carbon futures, the combined model of GJR-Normal1, Brent oil, coal, and electricity outperforms any other combinations. Fig. 4 shows the comparison of the forecasted values from model 12 and the realized volatility of carbon futures. In Fig. 5, we can compare the predictive accuracy of GJR-Normal1 and model 12. According to Fig. 5, the forecasted value of GJR-Normal1 is more dispersed from the realized volatility than that of model 12. Therefore, the combined model, which consists of GJR-Normal1 and energy variables, outperforms the single model of GJR-Normal1.

7. Conclusion

In this article, we have investigated the specification of different GARCH-type models to forecast carbon futures volatility. First of all, according to information criteria, the models with more complicated specifications of AR, MA, ARCH, and GARCH terms are rejected confirming previous literature (Benz and Trück, 2009). Secondly, through fitting to GARCH-type models, the residuals approach a normal distribution. Thirdly, the estimated parameters in the models are robust to the distribution assumed for the residuals. Finally, the asymmetric effect is observed in carbon futures, and this phenomena is also reported in Chevallier (2009, 2011b). When combining the above results and model selection criteria, we can conclude that the GJR-GARCH model performs better than other models.

When comparing models, the GARCH-type models perform better than implied volatility (IV) model, probably due to the low trading volume of carbon options. The GARCH-type models outperform the *k*-nearest neighbor model, because the process of carbon futures volatility

Table 13
Estimation results of Eq. (20), i.e., estimated parameters, *t*-statistics, information criteria, adjusted R^2 and *F* test, when the proxy for volatility is regressed on an intercept and the forecast obtained from the GJR-Normal1 and one to four proxies for the volatility of Brent oil, coal, natural gas, and electricity.

	α	$\beta_{Forecast}$	AIC	BIC	Adj-R^2	*F* test
Model 1	0.0019	0.8340***	−10.4596	−10.4385	0.5876	8.010×10^{-6}
	(1.4522)	(8.079)				
Model 2	0.0025*	0.8038***	−10.3681	−10.3470	0.5481	1.042×10^{-6}
	(1.9116)	(7.9411)				
Model 3	0.0024*	0.8026***	−10.3851	−10.3640	0.5557	4.029×10^{-7}
	(1.8762)	(7.9196)				
Model 4	0.0023*	0.8132***	−10.3770	−10.3559	0.5521	3.504×10^{-6}
	(1.7590)	(7.8602)				
Model 5	0.0021*	0.8243***	−10.4416	−10.4205	0.5801	4.042×10^{-6}
	(1.6769)	(8.2833)				
Model 6	0.0021	0.8261***	−10.4603	−10.4391	0.5878	2.732×10^{-6}
	(1.6230)	(8.3006)				
Model 7	0.0020	0.8340***	−10.4509	−10.4298	0.5840	1.421×10^{-5}
	(1.5277)	(8.1761)				
Model 8	0.0026**	0.7954***	−10.3638	−10.3427	0.5461	2.738×10^{-7}
	(2.0897)	(8.1614)				
Model 9	0.0025*	0.8046***	−10.3665	−10.3454	0.5473	1.320×10^{-6}
	(1.9410)	(8.0274)				
Model 10	0.0025*	0.8053***	−10.3739	−10.3528	0.5507	1.186×10^{-6}
	(1.9244)	(8.0236)				
Model 11	0.0022*	0.8162***	−10.4354	−10.4142	0.5775	1.216×10^{-6}
	(1.8354)	(8.4805)				
Model 12	0.0021*	0.8244***	−10.4384	−10.4172	0.5787	4.879×10^{-6}
	(1.7079)	(8.3359)				
Model 13	0.0021*	0.8267***	−10.4470	−10.4258	0.5823	5.756×10^{-6}
	(1.6738)	(8.3219)				
Model 14	0.0026**	0.7955***	−10.3579	−10.3368	0.5434	3.686×10^{-7}
	(2.1206)	(8.2076)				
Model 15	0.0023*	0.8158***	−10.4277	−10.4066	0.5742	1.580×10^{-6}
	(1.8675)	(8.4920)				
Reference model	0.0023*	0.8107***	−10.3840	−10.3628	0.5552	1.379×10^{-6}
	(1.6873)	(7.6874)				

Note: *** indicates significance at 1%, ** at 5%, and * at 10%. The *t*-statistics are in parentheses below corresponding parameter estimates. The column for *F* test indicates *p*-values for the unbiasedness of each model. The reference model is the forecast model that has the GJR-Normal1 as an explanatory variable only.

Table 12
MSE, MSE-LOG, MAE, MAE-LOG, and QLIKE from the GJR-GARCH with energy variables models.

	MSE		MSE-LOG		MAE		MAE-LOG		QLIKE	
	Value	Rank	Value	Rank	Value	Rank	Value	Rank	Value	Rank
Model 1	3.023×10^{-5}	1	0.1018	5	3.614×10^{-3}	7	0.2459	14	−3.238	15
Model 2	3.350×10^{-5}	13	0.1020	7	3.632×10^{-3}	12	0.2432	1	−3.248	6
Model 3	3.310×10^{-5}	11	0.1041	14	3.631×10^{-3}	11	0.2447	10	−3.241	14
Model 4	3.298×10^{-5}	10	0.1020	6	3.607×10^{-3}	5	0.2438	5	−3.246	9
Model 5	3.089×10^{-5}	5	0.1007	3	3.609×10^{-3}	6	0.2436	4	−3.246	10
Model 6	3.039×10^{-5}	2	0.1022	9	3.594×10^{-3}	3	0.2441	7	−3.241	13
Model 7	3.040×10^{-5}	3	0.1003	1	3.584×10^{-3}	1	0.2438	6	−3.245	12
Model 8	3.389×10^{-5}	15	0.1041	15	3.663×10^{-3}	15	0.2454	12	−3.247	7
Model 9	3.351×10^{-5}	14	0.1021	8	3.626×10^{-3}	9	0.2434	3	−3.250	1
Model 10	3.328×10^{-5}	12	0.1035	12	3.617×10^{-3}	8	0.2445	9	−3.248	3
Model 11	3.129×10^{-5}	7	0.1028	10	3.628×10^{-3}	10	0.2448	11	−3.246	11
Model 12	3.096×10^{-5}	6	0.1006	2	3.598×10^{-3}	4	0.2432	2	−3.248	4
Model 13	3.067×10^{-5}	4	0.1017	4	3.590×10^{-3}	2	0.2442	8	−3.247	8
Model 14	3.403×10^{-5}	16	0.1046	16	3.672×10^{-3}	16	0.2467	16	−3.250	2
Model 15	3.149×10^{-5}	8	0.1033	11	3.641×10^{-3}	14	0.2462	15	−3.248	5
Reference model	3.292×10^{-5}	9	0.1036	13	3.641×10^{-3}	13	0.2458	13	−3.238	16

Note: For each criterion, the first column shows the value of the criterion, the second column shows the rank of each model. The QLIKE value of the realized volatility is −3.367. Therefore, the rank in the QLIKE column is defined by how the QLIKE value of the forecasting model is different from that of the realized volatility. The reference model is the forecast model that has the GJR-Normal1 as an explanatory variable only.

Table 14
Diebold–Mariano test statistics, superior predictive ability test *p*-values, and model confidence set *p*-values for the GJR-GARCH with energy variables models.

	MSE		MSE-LOG		MAE		MAE-LOG		QLIKE	
Model 1	0.683	1.000**	−0.410	1.000**	−0.341	1.000**	−0.878	1.000**	−1.362	0.994**
Model 2	−1.665*	1.000**	−0.601	1.000**	−0.748	1.000**	7.568×10^{-4}	1.000**	0.112	1.000**
Model 3	−1.397	1.000**	−1.011	1.000**	−0.472	1.000**	−0.369	1.000**	−0.761	1.000**
Model 4	−1.259	1.000**	−0.546	1.000**	−0.168	1.000**	−0.200	1.000**	−0.067	1.000**
Model 5	0.155	1.000**	−0.052	1.000**	−0.482	1.000**	−0.229	1.000**	−0.523	1.000**
Model 6	0.672	1.000**	−0.539	1.000**	0.063	1.000**	−0.254	1.000**	−0.812	1.000**
Model 7	1.166	0.999**	0.241	1.000**	0.429	1.000**	−0.296	1.000**	−0.455	1.000**
Model 8	−1.791*	0.965**	−1.280	1.000**	−1.131	1.000**	−0.681	1.000**	−0.770	1.000**
Model 9	−1.525	1.000**	−0.715	1.000**	−0.668	1.000**	−0.101	1.000**	0.427	1.000**
Model 10	−1.366	1.000**	−0.984	1.000**	−0.298	1.000**	−0.360	1.000**	−0.320	1.000**
Model 11	−0.803	1.000**	−1.154	1.000**	−0.799	1.000**	−0.622	1.000**	−1.001	1.000**
Model 12	–	1.000**	–	1.000**	–	1.000**	–	1.000**	–	1.000**
Model 13	0.639	1.000**	−0.535	1.000**	0.175	1.000**	−0.338	1.000**	−0.568	1.000**
Model 14	−1.724*	1.000**	−1.577	0.979**	−1.360	0.975**	−1.172	0.997**	−1.077	1.000**
Model 15	−1.549	1.000**	−1.749*	1.000**	−1.378	1.000**	−1.435	1.000**	−1.673*	1.000**
Reference model	−1.293	0.999**	−0.872	1.000**	−0.712	1.000**	−0.718	1.000**	−1.023	1.000**
SPA_u	0.955		0.944		0.950		0.962		0.917	
SPA_c	0.955		0.944		0.950		0.962		0.917	
SPA_l	0.480		0.847		0.873		0.942		0.841	

Note: For each criterion, the first column shows the value of the Diebold–Mariano test statistics, and the second column shows the model confidence set *p*-values. * indicates significance at 10% in the first column. In the second column, ** indicates that the forecast model is included in $\widehat{M}^*_{75\%}$. The null hypotheses of these Diebold–Mariano statistics are that each model and model 12 have equal predictive accuracy. The benchmark model for the superior predictive ability test is the model 12.

possibly has a different pattern from the past movement, and is affected by some shocks from the exterior condition. In addition, the GJR-GARCH model with a normal distribution shows the best forecast performance among GARCH-type models. The GJR-GARCH model with a normal distribution outperforms the combination of two forecasts from IV, k-NN, and GARCH-type models. Therefore, the GJR-GARCH model with a normal distribution has the most information content about carbon futures volatility. When we introduce the volatilities of energy markets into the forecasting equation, Brent oil, coal, and electricity present a good predictive ability to forecast carbon futures volatility. As a result, to forecast the following day's carbon futures volatility, we can use the forecast from the GJR-GARCH model with the present day's volatilities of Brent oil, coal, and electricity.

This article has several important implications for both the risk management practices of firms that are involved in trading carbon futures and for investors. As noted in Daskalakis et al. (2009), the ability of market participants to speculate and hedge their positions in the EU ETS is important to maintain a high trading volume and to ensure market efficiency. Using the best forecast model of this article, market participants can hedge their positions by forecasting the volatility of carbon futures. Therefore, more sophisticated risk management for carbon futures market participants is available.

Moreover, with respect to the trading volume of carbon futures, we can state that the carbon futures market has a different characteristic from other financial markets. According to Donaldson and Kamstra (2005), GARCH-type forecasting models perform better than IV in the stock market during low-trading volume periods, and vice versa. From this result, Donaldson and Kamstra (2005) report that market prices contain more information relative to historical sources in high-trading volume periods than in low-trading volume periods. This result is inconsistent with the results of this article. Although the trading volume of carbon futures market is steadily growing, the predictability of the GARCH-type forecasting model is much better than that of IV. Therefore, the result of the carbon futures market seems to come from the

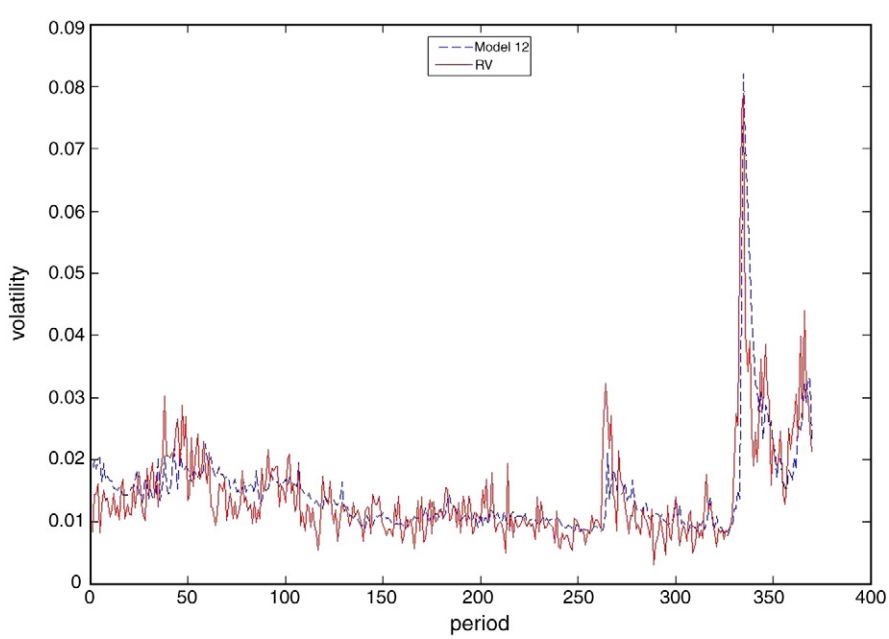

Fig. 4. The forecasted value from model 12, which consists of GJR-Normal1, Brent oil, coal, and electricity, and the realized volatility of carbon futures.

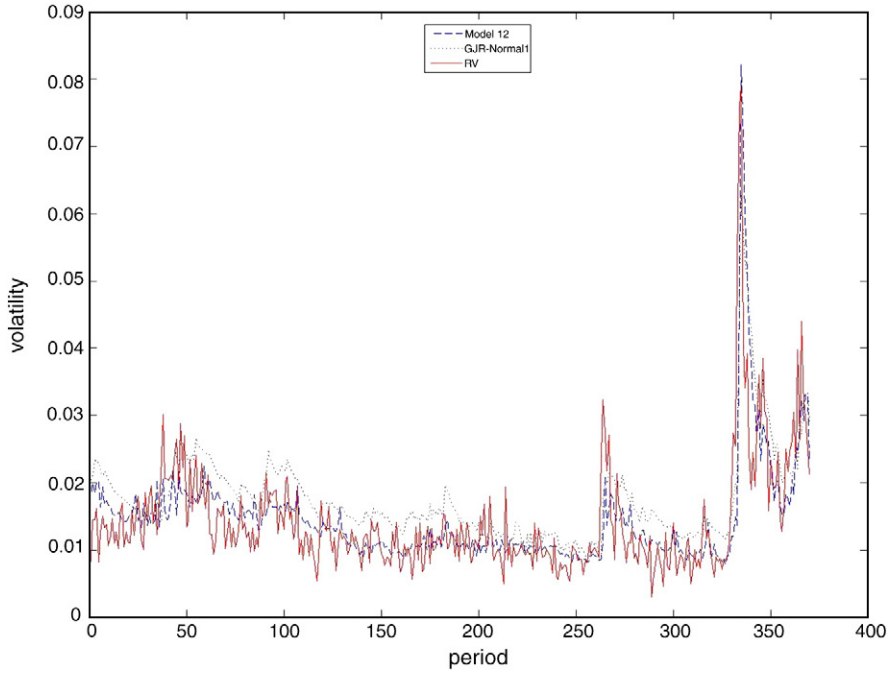

Fig. 5. Comparison of the forecasted values from GJR-Normal1 and model 12 with the realized volatility of carbon futures.

immaturity of the carbon option market. From this result, we can suggest that if the carbon option market becomes mature state, then the predictability of IV would be better.

With regard to the policy about the carbon market, we can make two suggestions from our results. The first suggestion is that the trading in the carbon option market should be invigorated so that the forecast from IV becomes more accurate as mentioned above. The other is that, when the policy makers formulate the policy about the carbon market, they had better consider the interaction between the carbon market and other energy markets, and set up the policy with the comprehensive and long-term view. Following Kossoy and Guigon (2012), it is definite that the policy about carbon market has an effect on the price of carbon futures market. They state that the oversupply of EUA and the recent economic recession of EU lead to the recent price drop of EUA. As a result, the policy for the EU ETS, which causes the oversupply of EUA, influences the downward tendency of the EUA price. Heimdal et al. (2012) and Kossoy and Guigon (2012) state that the intervention to elevate EUA price is needed, but this intervention should come into action with careful consideration and the long-term view.[19] Therefore, if the economic status becomes a normal state, and the allocation process of EUA is more elaborate, then the price of EUA would increase to the economically meaningful level. With this increase of the EUA price, the result of this article would be much more useful to forecast the volatility of carbon futures.

Acknowledgments

We thank two anonymous referees for their thoughtful and constructive comments and suggestions. This work was supported by the Human Resources Development program (No. 20094010200020-12-1-000) of the Korea Institute of Energy Technology Evaluation and Planning (KETEP) grant funded by the Korea government Ministry of Trade, Industry and Energy.

[19] This statement is consistent with Ellerman and Joskow (2008). They examine the carbon trading system in Phase I, and suggest that the European policy makers should take into account actual economic growth, other energy prices, and weather.

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